เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1 | Year: 2015

Subject: 341151 Electric Circuits I | **Section:** 5 - 6

Date: 13 October 2015 | **Time:** 15.00 - 17.00

Name:______________________ ID:________________ Class:______________

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Text books are **NOT** allowed.
4. Calculator is allowed.
5. The examination has 10 pages (including this page), 3 parts (59 questions) and a total score of 75 points.

Part 1 (15 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Engineering notation requires that all powers of ten be in multiples of six.
2. The prefix that corresponds to a power of ten of −9 is pico.
3. Early scientists came to the conclusion that like charges repel each other and unlike charges attract each other.
4. A potential difference is always measured at one point.
5. An ideal battery contains small amount of internal resistance.
6. A current source will supply, ideally, a fixed current even with changes in the terminal voltage of the device.
7. For good conductors, an increase in temperature will result in a decrease in the resistance level.
8. Conductance is the same as resistance.
9. Kirchhoff's voltage law requires that energy input to a system must equal the energy output from the system plus the energy lost or stored in the system.
10. The efficiency of a system is the ratio of the power output divided by the power input.
11. The current is the maximum through all elements of a series circuit.
12. The single-subscript notation V_a specifies the voltage at point a below ground.
13. The total resistance of parallel resistors is always less than the value of the smallest resistor.
14. For parallel elements of different value, the element with the smaller resistance will receive a smaller share of the current.
15. Series-parallel networks are networks that always contain twice as many series as parallel configurations.

Part 2 (40 points) Multiple choices

Instruction: Mark the correct answers for following questions in the provided answer sheet.

1. How many miles would a person walk in 3 h if his average pace is 15 min/mi?

   (a) 12 (b) 10

   (c) 14 (d) 16

2. 9×10^{4} = ____________ $\times 10^{7}$

   (a) 9×10^{-3} (b) 9×10^{4}

   (c) 9×10^{-4} (d) 9×10^{3}

3. Perform the indicated operation and express your answer in powers of ten: $(-0.002)^{2}$

   (a) 4×10^{6} (b) 2×10^{-6}

   (c) 2×10^{-6} (d) 4×10^{-6}

4. Convert 22.5 ms to microseconds.

   (a) 0.025 μs (b) 22,500μs

   (c) 22.5 μs (d) 0.000025μs

5. In double-subscript notation, the voltage V_{ab} is the same as

   (a) V_a-V_b (b) $V_a + V_b$

   (c) $V_a \times V_b$ (d) V_a

6. What current will a battery with an Ah rating of 500 theoretically provide for 30 h?

   (a) 17.7 A (b) 15.7 A

   (c) 16.7 A (d) 18.7 A

7. Calculate the area in circular mils (CM) of a wire having a diameter of 0.002 cm.

   (a) 0.62 CM (b) 5.25 CM

   (c) 6.2 CM (d) 53.5 CM

8. Find the conductance of a 0.65 Ωresistor.

   (a) 15.45 S (b) 0.154 S (c) 1.54 S (d) 1.54 S

9. If a freezer draws 1.5 A at 115 V, what is its internal resistance?

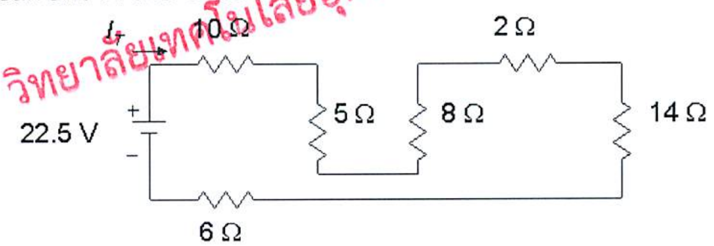
   (a) 7.7 Ω (b) 0.01Ω (c) 77Ω (d) 100Ω

10. In what range of values must a resistor exist if the color code is green, orange, and blue?
   (a) 42.4 MΩ through 63.6 MΩ
   (b) 52.94 MΩthrough 53.06 MΩ
   (c) 61.94 MΩthrough 64.06 MΩ
   (d) 41.94 MΩthrough 44.06 MΩ

11. The resistance of a light bulb changes as it heats. What is the resistance of a 120 V, 60 W bulb when it has been on for a few minutes?
   (a) 2400 Ω (b) 0.5Ω (c) 240Ω (d) 5Ω

12. What is the total cost of using an 860 W air conditioner for 24 h at a rate of 9 Baht per kilowatthour?
   (a) 1.86 Baht (b) 86 Baht (c) 18.6 Baht (d) 186 Baht

13. Find the total current in the circuit.

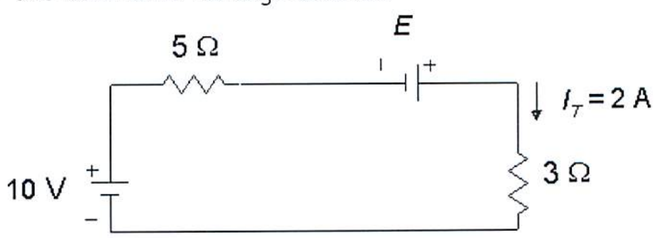


   (a) 0.5 A (b) 0.3 A (c) 0.7 A (d) 0.1 A

14. The double-subscript notation V_{ab} specifies point a as the point with __________ potential than point b.
   (a) the same (b) higher (c) different (d) lower

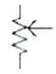
15. A physical voltage source _____________ have internal resistance.
   (a) cannot (b) may (c) will always (d) should

16. Find the value for the unknown voltage source.

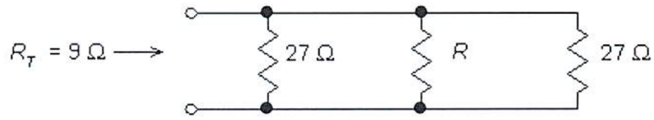


   (a) 8 V (b) 6 V (c) 4 V (d) 2 V

17. What is the symbol of potentiometer?

(a)  (b)  (c)  (d) 

18. Find the value of R.

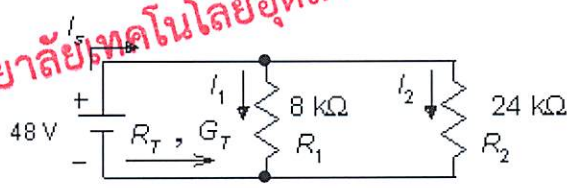


(a) 9Ω (b) 54Ω (c) 18Ω (d) 27Ω

19. How many joules of energy will a 10 Ωresistor connected across a 15 V battery dissipate in 1 min?

(a) 13.5 kJ (b) 135 kJ (c) 0.13 kJ (d) 1350 J

20. Find the total conductance.



(a) 166.7 μS (b) 1.667 S (c) 1.667 mS (d) 16.7 S

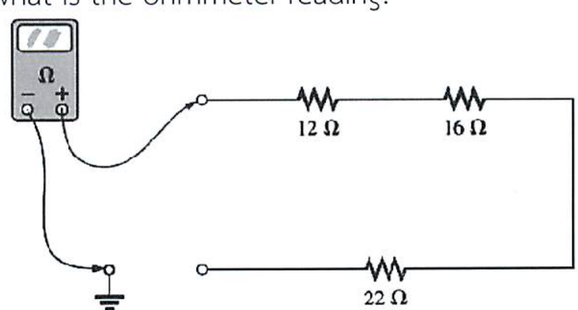
21. From circuit in problem 20, find the power dissipated by the resistor R_2.

(a) 960 mW (b) 96 W (c) 9.6 mW (d) 96 mW

22. What is the power deliver to a resistor 10kΩ with voltage across of 10 volts?

(a) 1 mW (b) 10 mW (c) 100 mW (d) 1 W

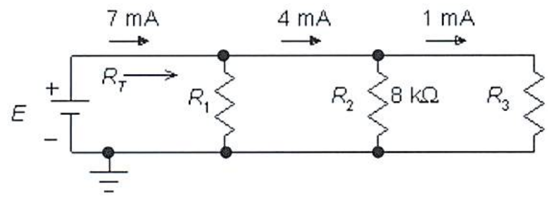
23. From figure below, what is the ohmmeter reading?



(a) 0 Ohms (b) 50 Ohms

(c) 12 Ohms (d) Infinity Ohms

24. Find R_1.



(a) 11 kΩ (b) 12 kΩ (c) 9 kΩ (d) 8 kΩ

25. From figure in problem 24, what is R_T ?

(a) 343Ω (b) 34.3 kΩ (c) 0.343 kΩ (d) 3.43 kΩ

26. For single-source parallel networks, the source current is ______________ the sum of the individual branch currents.

(a) smaller than (b) equal to

(c) double (d) more than double

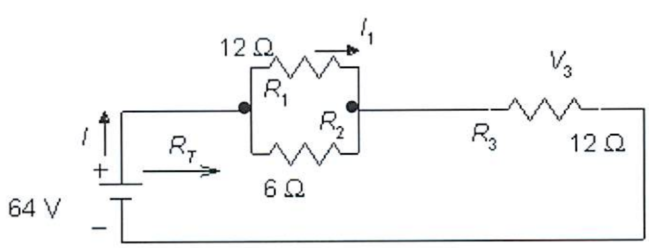
27. Another way to express Kirchhoff's current law is to say that the algebraic sum of all currents ______________ a junction equals zero.

(a) entering (b) through (c) leaving (d) at

28. Earth ground is a ground that is connected ______________ to the earth.

(a) partially (b) indirectly (c) directly (d) inadvertently

29. Calculate R_T.



(a) 30Ω (b) 16Ω (c) 24Ω (d) 18Ω

30. Use the circuit of Problem 29 and find the current I_1.

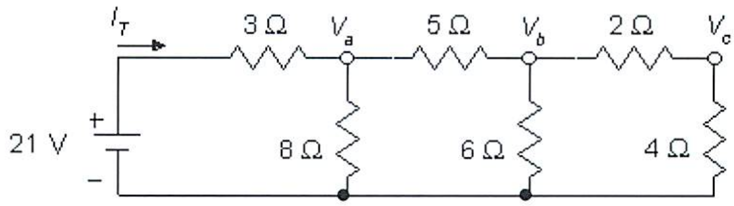
(a) 13.33 A (b) 1.333 A (c) 2.5 A (d) 0.25 A

31. A ladder network is named for ______________________________.

(a) its parallel structure (b) its series structure

(c) its inventor (Mr. Ladder) (d) its repetitive structure

32. Find I_T.



(a) 3 A (b) 3.5 A (c) 2.5 A (d) 4 A

33. Use the circuit of Problem 32 and find V_c.

(a) 4.5 V (b) 12 V (c) 6 V (d) 3 V

34. How much charge passes through a battery of 6.3 V if the energy expended is 80 J?

(a) 13.5 C (b) 12.7 C (c) 10.6 C (d) 11.4 C

35. The resistance between the two outside terminals of a potentiometer ___________.

(a) increases with the setting
(b) depends on the setting
(c) is always fixed
(d) decreases with the setting

36. If a resistor uses 420 J of energy for 7 min, what is the power to the resistor?

(a) 1 W (b) 80 mW (c) 80 W (d) 1 mW

37. A potential difference is always measured __________________.

(a) through a circuit
(b) around a circuit
(c) at one point
(d) between two points

38. The three major systems of units are the ________ system, the __________ system, and the _________ system.

(a) MKS, CGS, SI
(b) metric, MKS, CGS
(c) English, metric, SI
(d) English, French, German

39. The term "loaded" usually means the application of ____________________ to a supply so that it will draw current from the supply.

(a) a short
(b) network or system voltage
(c) an element
(d) an open

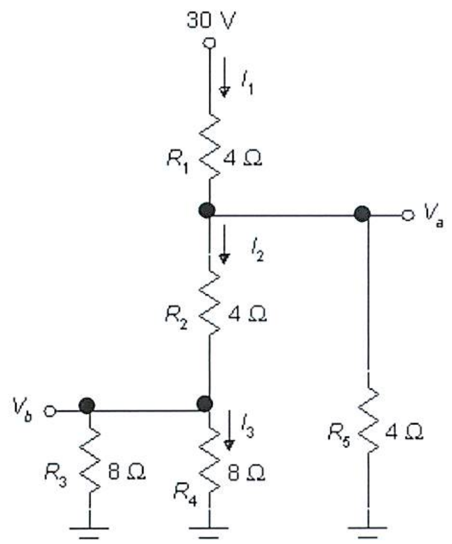
40. Insulators are materials that have ___________ free electrons and require a large applied potential to establish a measurable current level.

(a) several (b) very few (c) many (d) some

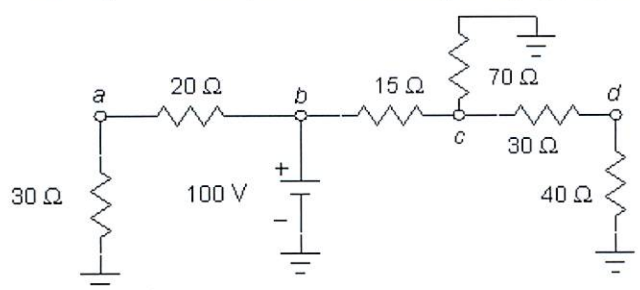
Part 3 (20 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

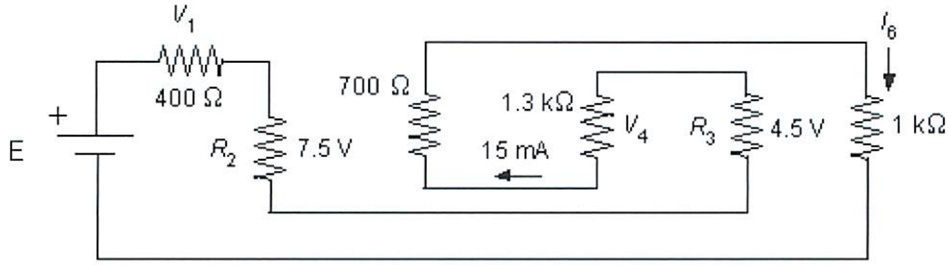
1. From Figure below, determine the value current I_1, I_2, I_3, $V_b$and V_a



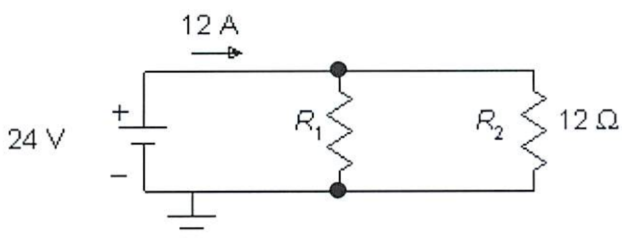
2. From Figure below, find the voltages V_a,V_b , V_c and V_d.



3. Find the value of E, R_2 and R_3



4. Determine the resistance R_1.



เลขที่นั่งสอบ

Answer Sheet

Name:______________________________ID:______________

Subject: 341151 Electric Circuits I

Part 1

	True	False
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Part 2

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